

```

// Ripple counter (See Fig. 6.8(b))
'timescale 1ns / 100 ps
module Ripple_Counter_4bit (A3, A2, A1, A0, Count, Reset);
    output A3, A2, A1, A0;
    input Count, Reset;
// Instantiate complementing flip-flop
    Comp_D_flip_flop F0 (A0, Count, Reset);
    Comp_D_flip_flop F1 (A1, A0, Reset);
    Comp_D_flip_flop F2 (A2, A1, Reset);
    Comp_D_flip_flop F3 (A3, A2, Reset);
endmodule
// Complementing flip-flop with delay
// Input to D flip-flop = Q'
module Comp_D_flip_flop (Q, CLK, Reset);
    output    Q;
    input     CLK, Reset;
    reg       Q;
    always @ (negedge CLK, posedge Reset)
    if (Reset) Q <= 1'b0;
    else Q <= #2 ~Q;          // intra-assignment delay
endmodule
// Stimulus for testing four-bit ripple counter
module t_Ripple_Counter_4bit;
    reg      Count;
    reg      Reset;
    wire     A0, A1, A2, A3;
// Instantiate ripple counter
    Ripple_Counter_4bit M0 (A3, A2, A1, A0, Count, Reset);
    always
    #5 Count = ~Count;
    initial
        begin
            Count = 1'b0;
            Reset = 1'b1;
            #4 Reset = 1'b0;
        end
    initial #170 $finish;
endmodule

```
